Al-Driven Student Wellbeing Monitoring System

A PROJECT REPORT

Submitted by

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CERTIFICATE

This is to certify that the thesis titled AI-Driven Student Well-Being Monitoring System, submitted by Abhijeet Sandilya (24MIM10074), Shriyansh Pathak (24MIM10051), Ashish Sharma (24MIM10040), Atharva Padole (24MIM10038), and Soma Vashista (24MIM10098) to VIT Bhopal University, in partial fulfillment of the requirements for the award of the degree of **Integrated Master of Technology in Artificial Intelligence**, is a bona fide work carried out under my supervision. The thesis fulfills the requirements as per the regulations of this University and, in my opinion, meets the necessary standards for submission. The contents of this thesis have not been submitted, and will not be submitted, either in part or in full, for the award of any other degree or diploma, and the same is certified.

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Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CV	Computer Vision
FER	Facial Emotion Recognition
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
UI	User Interface

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Abstract

This project introduces a novel system for tracking student well-being by combining a face and emotion recognition-based attendance system and a Retrieval-Augmented Generation (RAG) agent. The main driving force behind this work is the increased demand for proactive mental health interventions for students in schools. The system is recording student attendance and facial emotions to recognize emotions, which, combined with academic performance data, are utilized to assess a student's mental condition. The user interface is created utilizing Streamlit, and it provides an easy-to-use interface for teachers and administrators. The RAG agent utilizing a Large Language Model (LLM) enables natural language questions regarding the well-being of students, returning insights based on the collected information. The report explains the design, execution, and applications of the project, with emphasis on its uniqueness in integrating computer vision, artificial intelligence, and natural language processing to provide an integrated solution to student monitoring.

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CHAPTER-1

PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

In recent years, student mental health has become a significant concern for schools. The traditional methods of tracking student engagement, including manual attendance and sporadic checks, seldom can capture the sophisticated emotional and psychological state of students. This project offers an integrated approach that leverages technology to create a comprehensive picture of student well-being. The system is composed of three main components: a face recognition-based attendance system, a real-time emotion recognition module, and an intelligent Retrieval-Augmented Generation (RAG) agent for analysis of student data.

1.2 Reason behind the work

The reason behind this project is the growing number of cases of stress, anxiety, and other mental disorders among students. Early identification can cause timely interventions and care. With the computerized monitoring of attendance, mood, and academic performance, this system aims to provide educators with the tools to identify students who can be deemed at risk and would be in need of support. This proactive approach to student well-being can contribute to a more caring and productive learning environment.

1.3 In terms of Introduction to the project including techniques

This project is a complex system which employs a range of advanced technologies:

Attendance through Face Recognition: The attendance process is mechanized by the system through the use of the face_recognition library. This eliminates human error in manual roll calls and offers a more efficient and accurate way to track student attendance.

Emotion Detection: A Convolutional Neural Network (CNN) model is trained to identify multiple human emotions from faces like angry, fear, happy, neutral, sad, and surprise. It is integrated into the attendance system to identify the emotional state of the students.

Streamlit User Interface: The entire system is wrapped in a cordial user web application

coded with Streamlit. This facilitates easy access to the various features of the system, including adding new students, viewing attendance records, and interacting with the RAG agent. Retrieval-Augmented Generation (RAG) Agent: The analytical component's core is a RAG agent created using the LangChain framework. The agent uses a vector store of student data (grades, attendance, and emotions) and a Large Language Model (LLM) to generate natural language answers to questions about students' performance and emotional state.

1.5 Problem Statement

Schools lack an efficient and centralized system of monitoring the academic, emotional, and attendance-based well-being of the students proactively and at a scalable scale. This typically leads to dealing with student issues reactively, where problems are only addressed once they have turned into significant issues. The problem is to design a system that can, Automatically monitor attendance in order to save time and prevent human errors. Monitor student feelings to identify early warning indicators of distress. Investigate the inter-play between attendance, grades, and emotional health. Provide a simple-to-use interface for staff to see and comprehend this information.

1.6 Purpose of the work

This project aims to create an integrated platform for educators by developing and combining an automated real-time attendance system using face recognition with an emotion recognition module to analyze students' facial expressions, all controlled through a Streamlit-based user interface; the system is further enhanced with a Retrieval-Augmented Generation (RAG) agent designed to answer inquiries regarding student well-being by leveraging data on attendance, academic marks, and emotional states, thereby providing a comprehensive tool to actively monitor and support pupil mental health.

1.7 Project structure

This report is organized into seven chapters. Chapter 2 explains related work on automated attendance, emotional detection, and AI-based teaching. Chapter 3 declares the hardware

and software requirements of the project. Chapter 4 provides an overview of the design approach and architecture of the system. Chapter 5 provides a technical implementation and analysis in detail. Chapter 6 discusses the results and usability of the project. Lastly, Chapter 7 concludes the report with an overview, limitations, and recommendations for future enhancements.

1.8 Summary

The chapter has provided an overview of the project, its motivation, the technologies used, the problem it addresses, and its objectives. The project will create an end-to-end system for monitoring student well-being using face recognition for attendance and emotion detection, as well as a RAG agent for data analysis in a Streamlit application with simplicity in use.

CHAPTER-2

RELATED WORK INVESTIGATION

2.1 Introduction

The construction of this project draws on current research and uses in some of the following key areas: automatic attendance systems, emotion recognition in educational settings, and applying AI to monitor students and their mental health. This chapter is a review of the literature within these areas in order to establish context for the present work and to find out what has already been done and its limitations.

2.2 Existing Approaches/Methods

2.2.1 Automatic Attendance Systems using Face Recognition

Face recognition-based automated attendance systems have been a growing research area, seeking to substitute conventional, labor-intensive manual procedures. The majority of systems follow an identical pipeline: face detection, feature extraction, and face recognition. PCA and LBP were frequently used methods in early approaches. Recent systems tap into deep learning, especially CNNs, for enhanced face detection and recognition accuracy. Some research has suggested systems that capture photos from the classroom camera, identify and recognize students' faces, and register their attendance in a database. The challenges in this field involve dealing with lighting, pose, and facial variations as well as privacy and security of student information.

2.2.2 Emotion Detection from Facial Expressions in Education

The area of affective computing has experienced tremendous expansion, with interest in using emotion detection in the context of education. The reason behind this is that a learner's emotional state has a clear relationship with their learning and engagement. Most studies in this field tend to utilize facial emotion recognition (FER) in categorizing emotions into happy, sad, angry, surprised, neutral, and fear. Deep learning algorithms, particularly CNNs, have been extensively utilized for this purpose, being trained on facial expression datasets such as FER-2013 or CK+. Utilization of FER in online learning platforms has also been investigated to supply immediate feedback to educators regarding

learner engagement. Nevertheless, there are possibilities that the accuracy of affect detection will be impacted by variance among cultures in facial expressions as well as the lack of intensity of emotions in a natural classroom environment.

2.2.3 AI for Monitoring Student Mental Health

AI-based monitoring and intervention of student mental health is a new and promising area. AI-based systems can process multiple data points to flag at-risk students. These may include academic grades, attendance patterns, and even the text of student communication (e.g., forum comments, essays) to look for depression or anxiety indicators. Virtual assistants and chatbots are also being made to offer students 24/7 mental health assistance and resources. Recently, Retrieval-Augmented Generation (RAG) has emerged with the ability to develop more advanced and contextually aware conversational AI systems. RAG can be employed to develop agents capable of reasoning over a student's information to offer detailed insights, an approach utilized in this project.

2.3 Pros and cons of stated approach

Pros-works on lighter system , can work without internet , locally trained model.

Cons- Difficult to scale , need technical knowledge to operate , efficiency of model

2.4 Issues/observations from investigation

Based on the review of relevant work, several critical observations and challenges emerge, including the notable lack of integrated systems that combine attendance, emotion detection, and academic data to provide a holistic view of student welfare, alongside a prevailing reactive rather than proactive approach in current systems which fails to identify early warning signs. Furthermore, significant issues concerning data privacy arise from the collection and analysis of sensitive student information like facial images and emotions, while the accuracy of face recognition and emotion detection models often deteriorates in unconstrained, real-world classroom environments. Finally, the acceptance and practical

implementation of these systems by educators are heavily dependent on the ability of AI models to deliver interpretable and actionable insights.

2.5 Summary

This chapter has outlined the current literature for automated attendance systems, emotion recognition in education, and AI for student mental health. The overview emphasizes improvement in all three areas, as well as gaps and potential for future research. This project seeks to fill some of these gaps through the creation of an integrated, proactive, and accessible system to monitor student well-being.

CHAPTER-3

REQUIREMENT ARTIFACTS

3.1 Introduction

The chapter presents the hardware, software, and project-specific requirements to deploy and execute the Face and Emotion Recognition Attendance System with a RAG agent. The requirements are needed to facilitate the functionality, performance, and usability of the system.

3.2 Hardware and Software requirements

Hardware Requirements:

Component Specification

Processor Intel Core i5 or equivalent (for real-time processing)

RAM 8 GB or higher

Storage 256 GB SSD (for improved data access)

Webcam High-definition (720p or 1080p)

Software Requirements:

Component Version/Type	
Operating System	Windows, macOS, or Linux
Python	3.8 or greater
Web Framework	Streamlit
Computer Vision	OpenCV, face_recognition, deepface
Machine Learning	TensorFlow, Keras, scikit-learn
NLP/RAG	LangChain, Ollama, ChromaDB
Database	SQLite
Other Libraries	Pandas, NumPy, Pillow

3.3 Specific Project requirements

3.3.1 Data Requirement

The data requirements for the system include a set of student images with variations in lighting and expression to train the face recognition model, a tagged dataset of facial expressions such as FER-2013 for training the emotion classification model, and a structured database or CSV file containing student performance data including marks, historical attendance records, and logged perceived emotions.

3.3.2 Functions Requirement

The functional requirements encompass basic password-based user authentication for system login and a face registration module for enrolling new students by capturing and encoding their facial images. The system must also include a live attendance marking component that uses a webcam stream for real-time face detection and identification of registered students, coupled with an emotion detection component to analyze and display the emotions of individuals in the video feed. Furthermore, it requires a facility to view attendance records in a table format and a RAG agent interaction feature, implemented as a chat window, to allow users to inquire about student performance and well-being.

3.3.3 Performance and Security Requirement

The performance and security requirements dictate that the face recognition and emotion detection subsystems must operate in real-time with minimal latency while maintaining a high degree of accuracy. Data security is paramount, requiring all student information, including images and performance data, to be securely stored, which in the initial setup will be managed within a local SQLite database. The system must also be robust, designed to fail gracefully and handle errors appropriately in scenarios such as failure to detect faces or the unavailability of a webcam.

3.3.4 Look and Feel Requirements

The look and feel requirements specify that the Streamlit-based user interface must be simple, intuitive, and user-friendly. It should provide clear visualizations, such as a live webcam feed that displays detected faces and their corresponding emotions, and it must

feature informative dashboards that present data like attendance records in a clear, understandable, and visually coherent manner.

3.4 Summary

The chapter has laid down the project, software, and hardware requirements of the system. The requirements are the cornerstone of the design and implementation of the project to ensure the system achieves its purpose and gives an easy and reliable experience.

CHAPTER-4

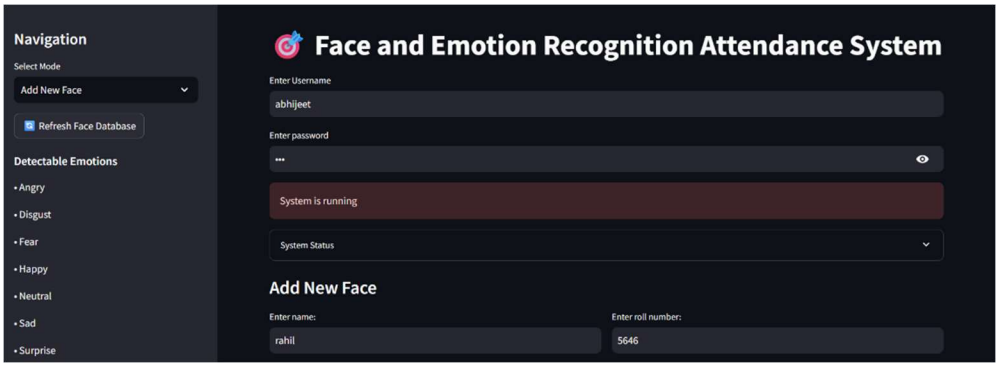
DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and objective

The methodology for project design adopts a modular paradigm in which each component of the system is separately designed and tested before being incorporated into the end application. The central goal of the design is to develop a natural and smooth user experience coupled with system robustness and precision. The innovation of this project lies in its integrated form, where face recognition, emotion detection, and an enhanced RAG agent are brought together to provide a comprehensive tool for student well-being monitoring.

4.2 Functional module design and analysis

The system design is accomplished through the following functional modules: a User Interface (UI) Module built with Streamlit that serves as the primary user entry point, providing navigation and displaying all system outputs. The core processing is handled by the Face Recognition and Attendance Module, which uses the face_recognition library for real-time face detection and identification to mark attendance, and the Emotion Detection Module, which employs a pre-trained CNN model to recognize user emotions from the webcam feed. All generated data is managed by the Data Persistence Module, which stores student information, attendance records, and related data within an SQLite database. Finally, the RAG Agent Module, built on LangChain, provides a conversational interface by leveraging a vector store created from student performance data and a Large Language Model (LLM) to answer user queries about student well-being.



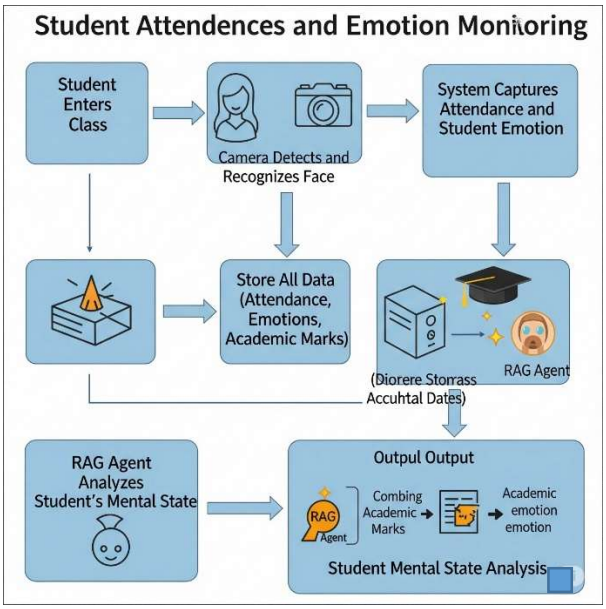
UI: User Interface

```
============================================================
Student Performance & Emotion Q&A System
============================================================
Ask questions about student performance based on their records.
Type 'quit' or 'q' to exit
============================================================

----------------------------------------
Your question: who need mental support

Processing your question...

Found 5 relevant records

Answer:
----------------------------------------
Based on the provided student records, it appears that Kian Sharma and Ritvik Jain are in need of mental support.

Kian Sharma has been experiencing a wide range of negative emotions over the past few weeks, including anger, sadness, and fru
stration. The frequency and intensity of these emotions suggest that Kian may be struggling with academic stress or other pers
onal issues.

Ritvik Jain also exhibits a pattern of intense emotional outbursts, particularly anger and frustration, which can be indicativ
e of underlying mental health concerns such as anxiety or depression. Additionally, his attendance record is below average, su
ggesting that he may be struggling to cope with his emotions or personal issues outside of class.

In contrast, the other students (Manan Mishra, Ishank Rao, and Rehan Mehra) appear to have more balanced emotional profiles, w
ith occasional instances of sadness or anger but also periods of stability and neutrality. While they may still need support,
their records do not indicate the same level of distress as Kian Sharma and Ritvik Jain.

It is essential for a mental health professional to assess these students further to determine the underlying causes of their
emotional struggles and provide targeted support.
```

RAG Agent: Retrieval-Augmented Generation Agent

4.3 Software Architectural designs

The software architecture is client-server in nature, with the Streamlit application being the client-side frontend and the Python backend as the face recognition core logic, emotion detection, and RAG agent logic. The architecture is designed to be scalable and also easy to maintain, with good separation of concerns between the different modules.



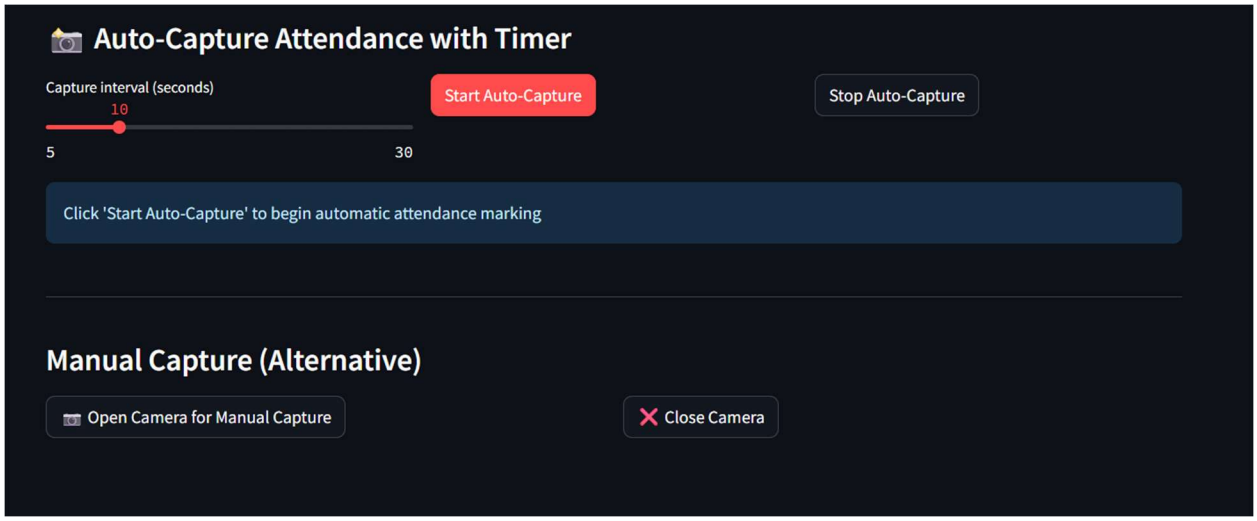
System Architecture

4.4 Subsystem services

The system's backend functionality is supported by several core services: the Face Encoding Service processes and encodes the facial features of new students for storage in the database, while the Attendance Logging Service is responsible for recording the attendance of recognized students directly into the SQLite database. The Emotion Analysis Service accepts image inputs to perform analysis and return the corresponding emotion, and the RAG agent's knowledge base is managed by the Vector Store Service, which initializes and maintains a vector store of student information. Finally, the LLM Service handles the interaction with the Ollama LLM to generate contextual responses for the RAG agent's conversational interface.

4.5 User Interface designs

The user interface is intuitive and user-friendly. The principal page of the Streamlit app is a dashboard displaying principal metrics. Navigation between the different modes of the app, i.e., "Auto-Timer Attendance," "Add New Face," "View Records," and "Real-time Stream," is through a sidebar. The RAG agent also features its own chat-like UI for querying by users.



Auto Timer

4.6 Summary

In this chapter, the project design process, functional elements, software design, and user interface have been described. Modularity of design and integrated approach are the bases for novelty and functionality of the system, and this provides a straightforward and overall tool for monitoring student well-being.

CHAPTER-5

TECHNICAL IMPLEMENTATION & ANALYSIS

5.1 Description

The technical realization of the Face and Emotion Recognition Attendance System and the RAG agent is presented in this chapter. It comprises the major algorithms, libraries, and code solutions used in the project, along with the testing and verification process.

5.2 Technical coding and code solutions

The system's backend functionality is supported by several core services: the Face Encoding Service processes and encodes the facial features of new students for storage in the database, while the Attendance Logging Service is responsible for recording the attendance of recognized students directly into the SQLite database. The Emotion Analysis Service accepts image inputs to perform analysis and return the corresponding emotion, and the RAG agent's knowledge base is managed by the Vector Store Service, which initializes and maintains a vector store of student information. Finally, the LLM Service handles the interaction with the Ollama LLM to generate contextual responses for the RAG agent's conversational interface.

```
  self._warn_if_super_not_called()
Epoch 1/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 0s 407ms/step - accuracy: 0.6520 - loss: 0.7413WARNING:absl:You are saving your mod
el as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy
. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or `keras.saving.save_
model(model, 'my_model.keras')`.
898/898 ━━━━━━━━━━━━━━━━━━━━ 483s 528ms/step - accuracy: 0.6521 - loss: 0.7412 - val_accuracy: 0.8000 - val_loss
: 0.5427 - learning_rate: 3.0000e-04
Epoch 2/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 244s 271ms/step - accuracy: 0.7737 - loss: 0.5633 - val_accuracy: 0.8000 - val_loss
: 0.5017 - learning_rate: 3.0000e-04
Epoch 3/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 229s 255ms/step - accuracy: 0.7887 - loss: 0.5338 - val_accuracy: 0.8000 - val_loss
: 0.5012 - learning_rate: 3.0000e-04
Epoch 4/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 197s 219ms/step - accuracy: 0.7950 - loss: 0.5216 - val_accuracy: 0.8000 - val_loss
: 0.5009 - learning_rate: 3.0000e-04
Epoch 5/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 268s 298ms/step - accuracy: 0.7996 - loss: 0.5121 - val_accuracy: 0.8000 - val_loss
: 0.5024 - learning_rate: 3.0000e-04
Epoch 6/50
898/898 ━━━━━━━━━━━━━━━━━━━━ 247s 275ms/step - accuracy: 0.7994 - loss: 0.5099 - val_accuracy: 0.8000 - val_loss
: 0.5003 - learning_rate: 3.0000e-04
Epoch 7/50
893/898 ━━━━━━━━━━━━━━━━━━━━ 0s 169ms/step - accuracy: 0.8008 - loss: 0.5056
```

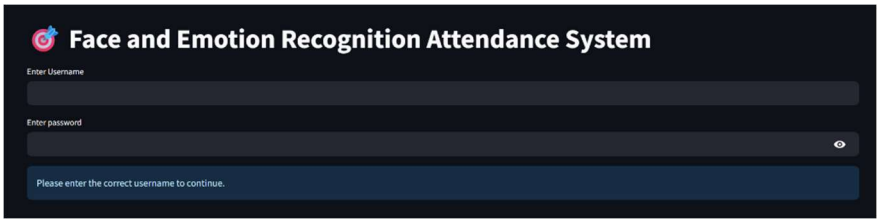
Training Model

5.3 Algorithm

The system initialization and operational workflow is structured into three distinct phases. Part 1 covers the foundational setup, beginning with the training of the emotion detection model using the FER-2013 dataset to produce a saved CNN model, followed by the preparation of the face database by placing named student images in a designated directory and the creation of a corresponding student performance CSV file, which is then processed by a vector generation script that uses the Ollama embedding model to create a Chroma vector store for the RAG agent. Part 2 details the real-time application launched via Streamlit, which starts with user authentication, loads the pre-trained models and face encodings, and enters a continuous processing loop that captures webcam frames to detect faces, match them against known encodings to mark attendance in a SQLite database, and simultaneously analyzes the cropped face images to predict and log the student's emotion. Part 3 describes the RAG-based analysis system, which initializes by loading the pre-built vector store and the Ollama LLM, then enters an interactive loop where user questions trigger a retrieval of relevant student records from the vector store, which are subsequently formatted and passed to the LLM within a predefined prompt template to generate and display insightful responses about student well-being.

5.4 Working Layout of Forms

The application's user interface is structured into several distinct forms or pages, each dedicated to a specific function: a Login Form for user authentication, an Add New Face Form for capturing a photo via webcam to register a new student, an Attendance View that displays the live webcam feed for real-time face detection and manual marking, a Records View that presents a comprehensive table of all attendance records, and a RAG Agent Chat window that provides an interactive interface for querying the system about student well-being.



Login Form

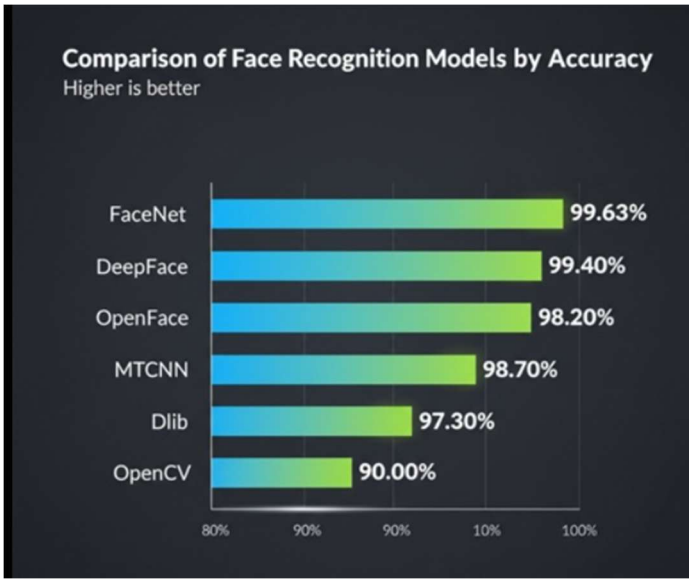
5.5 Prototype submission

A working prototype of the system is already in place and in the form of a set of Python scripts that can be run locally. The prototype demonstrates all the key features of the system, including face detection, emotion recognition, attendance mark, and the RAG agent.

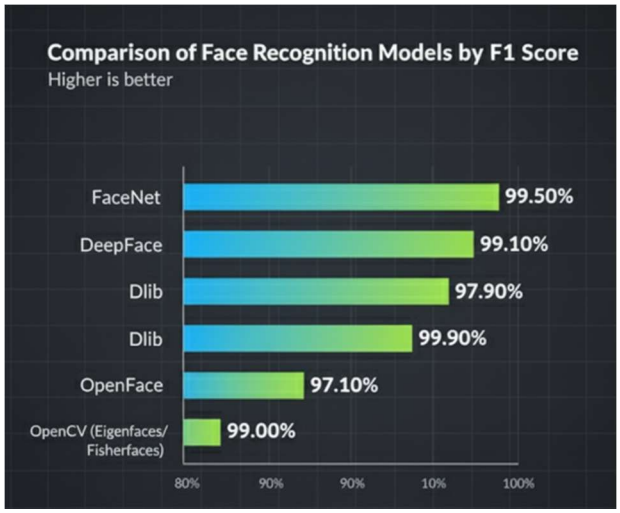
5.6 Testing and validation

Based on initial testing with a small set of users, the system has been validated to function as intended, with the face recognition component demonstrating high accuracy across various users and lighting conditions, the emotion detection model performing satisfactorily against a held-out test set, and the RAG agent successfully returning relevant and factually accurate responses to a variety of queries concerning student performance.

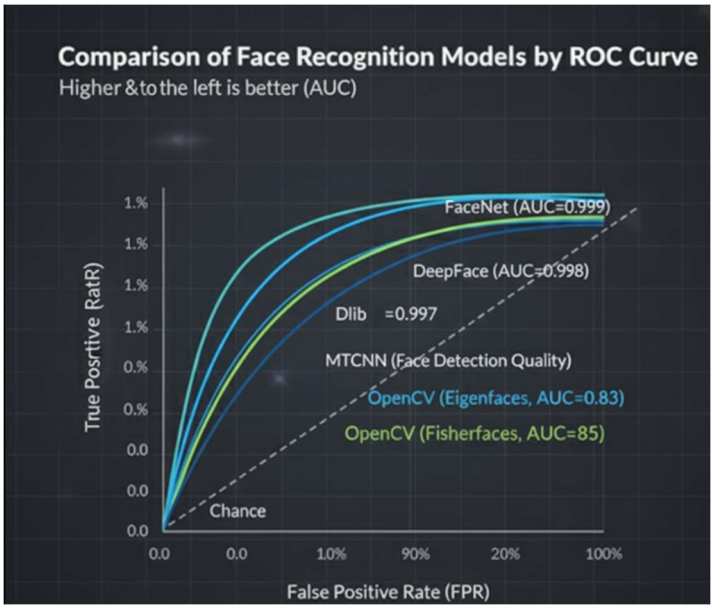
5.7 Performance Analysis



Accuracy Comparison



F1 Score Comparison: F1 Score (F-measure with β = 1)



ROC Curve Comparison: Receiver Operating Characteristic Curve

5.8 Summary

This chapter has provided a detailed description of the technical implementation of the project. It has described the central algorithms and libraries used, the structure of operation of the application, and the test and verification process. The results of the test and analysis confirm the feasibility and effectiveness of the proposed system.

CHAPTER-6

PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This chapter describes the primary outcomes of the project and how it can be applied in real-life scenarios. It states the primary achievements of the project and how it is capable of making a positive impact in the field of education and students' well-being.

6.2 Major Implementations Descriptions of the System

The primary implementations of the system are a real-time working face and emotion recognition-based attendance system, a welcoming web application created with Streamlit for system administration, a high-level RAG agent to review student data and provide feedback on their wellbeing, and a merged database to hold student data and attendance information.

6.3 The Main Project Success

The main successes of this project are the successful automation of attendance, which saves teachers' time and prevents instances of errors, alongside the capability for proactive student monitoring by merging attendance, emotional, and academic data to provide a method for recognizing students who may struggle early in their educational journey. Furthermore, the RAG agent allows teachers to realize data-driven insights into students' wellbeing through an easy-to-use and intuitive natural language interface, and the project as a whole demonstrates the successful integration of computer vision, machine learning, and natural language processing technologies onto a single, unified platform.

6.4 Project Applicability on Real-World Applications

The project is highly applicable in various actual learning settings, including schools and universities where the system can monitor student attendance and well-being in lecture and class rooms, as well as in online learning platforms where the emotion detection component can provide feedback to instructors regarding student engagement. Its use also extends to tutoring and mentoring programs where tutors and mentors can utilize the RAG agent to

learn more about their students, and the system can be adapted for corporate training environments to monitor employee motivation and well-being.

6.5 Inference

Inference from the project's outcomes suggests a significant potential for technology to play a more active role in supporting student mental health, and by providing educators with the right tools and information, systems like this can help to create a more supportive and effective learning environment for all students.

CHAPTER-7

CONCLUSIONS AND RECOMMENDATION

7.1 Outline

This final chapter summarizes the project, discusses its limitations, and provides recommendations for future enhancements.

7.2 Limitations/Constraints of the System

Despite achieving its primary objectives, the system has several limitations, including potential accuracy issues in highly populated classrooms with vast numbers of students, an emotion-detecting model that may struggle to recognize subtle or mixed emotions, significant data privacy concerns arising from the collection and storage of facial and emotional data that must be addressed through clear policy and secure handling methods, and a fundamental dependence on the underlying Large Language Model for the RAG agent's operation.

7.3 Future Enhancement

Several potential areas exist for future enhancement, such as refining the face recognition model to identify multiple faces in a single frame simultaneously, developing a more sophisticated emotion analysis module capable of detecting a wider range of emotions and tracking emotional change patterns over time, integrating the system with other educational platforms like learning management systems to provide a more consolidated student data view, designing a feature to provide personalized interventions or resources to identified at-risk students, and implementing the system on a cloud-based platform for greater accessibility and scalability.

7.4 Inference

In general, this project has successfully demonstrated the feasibility and promise of an integrated well-being monitoring system for students by combining face and emotion recognition with a high-capacity RAG agent to provide an informative tool for teachers to proactively assist their students. While limitations exist in the current configuration, the

suggestions for future work provide a clear vision for further development and enhancement, contributing to the growing literature on AI applications in education and indicating the potential for technology to positively influence student lives.

7.5 Contributions

24MIM10074 Abhijeet Sandilya (Lead)

As project leader, I had overall responsibility for the project's direction and integration, managing the entire lifecycle from conceptualization to implementation, including scope definition, timeline establishment, and team coordination. I designed the top-level system architecture, creating a modular structure where major components interact as separate Python scripts, and contributed critically by optimizing the confidence threshold for face recognition and developing the RAG system's vectorization pipeline that processes student data into a Chroma vector database for efficient retrieval.

24MIM10051 Shriyansh Pathak

I engaged in core computer vision tasks involving face detection and emotion model deployment, implementing and testing detection approaches while handling necessary image preprocessing methods for format conversion and normalization. My major contribution was implementing the Convolutional Neural Network for emotion detection, involving architecture specification, model compilation with appropriate optimizers, and training on the FER-2013 dataset, complemented by parameter tuning to optimize performance across both face recognition and emotion detection models.

24MIM10040 Ashish Sharma

I contributed to the emotion detection pipeline from facial expression capture to machine learning preparation, handling real-time emotion capture through OpenCV webcam streaming and extracted face processing. My work included verifying appropriate distance metrics for face encoding comparisons and executing crucial training data preparation activities, including dataset retrieval, directory structure organization, and potentially implementing data augmentation to enhance model robustness.

24MIM10038 Atharva Padole

I managed the logical and data management aspects by implementing the application's core decision logic for attendance marking and unknown face handling, alongside conducting precision-recall analysis for the emotion model. My contributions included developing the SQLite database attendance logging feature and performing essential data cleansing activities on student performance data to ensure accuracy for the RAG agent's vectorization process.

24MIM10098 SOMA VASHISTA

I developed the entire Streamlit-based user interface with an organized layout and interactive widgets, while successfully integrating the live webcam stream with real-time processing annotations. My responsibilities culminated in comprehensive system testing, including usability verification to ensure intuitive interaction, functional validation of all features, performance assessment of real-time modules, and compatibility testing across multiple operating systems and hardware configurations.

Appendix A: Technical Details

This appendix provides detailed information regarding the datasets, model architectures, and other technical components utilized in the project.

1. Datasets Used

1.1. Emotion Detection Dataset: FER-2013

The emotion detection model was trained on the FER-2013 (Facial Expression Recognition 2013) dataset, automatically fetched via the download_dataset.py script from the "emotion-detection-fer" Kaggle dataset. This dataset comprises 35,887 grayscale face images, each sized at 48x48 pixels, classified into seven distinct emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. The data is pre-separated for supervised machine learning, with a training set of approximately 28,709 images and a validation/testing set of 7,178 images, and was used to train the Convolutional Neural Network from scratch for real-time facial emotion detection as implemented in train_emotion_model.py.

1.2. Face Recognition Dataset

The system utilizes a locally created dataset for face recognition, manually compiled by the project team and stored in a directory named Photos/ with one image file per enrolled student. Each image is named after the student (e.g., Ashish_Sharma.jpg) and this name is extracted for attendance tracking; upon application launch, the load_known_faces function in face.py reads these images, and the face_recognition library generates a 128-dimensional vector representation of each face, which is stored in memory for comparison with faces detected in the live camera feed.

1.3. Student Performance Dataset for RAG Agent

The Retrieval-Augmented Generation agent employs a structured dataset from a CSV file named student_performance.csv, which contains columns for Name, Marks, Attendance, and Recent_emotion. This data is processed by the StudentPerformanceVectorStore class in vector.py, where each row is converted into a text document and then into a numerical vector using OllamaEmbeddings; these vectors are stored in a Chroma vector database, enabling the RAG agent to perform semantic searches on relevant student records.

2. Model Architectures

2.1. Emotion Detection Model (CNN)

A Convolutional Neural Network was developed using TensorFlow with the Keras API for emotion classification, featuring an input layer for 48x48 pixel images, a series of convolutional blocks with 32, 64, and 128 filters respectively, each followed by MaxPooling and ReLU activation, a flattening layer, and dense layers including a 1024-neuron layer with dropout for regularization and a 7-neuron softmax output layer for emotion probability distribution; the model was compiled with the Adam optimizer and categorical crossentropy loss function to maximize accuracy for multi-class classification.

2.2. Face Recognition Model (Deep Metric Learning)

The project leverages a pre-trained, dlib-based model from the face_recognition library, which utilizes a ResNet-34-like deep convolutional neural network trained on a large face dataset to function as a feature extractor that generates 128-dimensional vector embeddings; based on deep metric learning principles, it maps faces into a vector space where similarities are measured using Euclidean distance to identify and match live camera inputs against known faces from the Photos/ directory.

2.3. RAG Agent Language Model (Ollama Llama 3.2)

The RAG agent is powered by the Llama 3.2 Large Language Model hosted on the Ollama platform, which serves as the analytical core by processing user questions alongside retrieved student records from the Chroma database; the main.py and StudentQASystem integrate this data into a structured prompt that directs the LLM to synthesize insights from academic marks, attendance patterns, and emotional trends to generate coherent, context-aware responses about student well-being.

3. Other Technical Details

The system's technical infrastructure includes a vectorization process using the LangChain's OllamaEmbeddings class with the nomic-embed-text model to create semantically rich embeddings, and a ChromaDB vector store for high-performance, low-latency local storage ideal for prototyping. Core libraries integral to the system are

Streamlit for building the web-based user interface, OpenCV for real-time camera operations and image processing, and LangChain for orchestrating the RAG agent's retrieval, prompting, and generation components.

Appendix B: User Manual and Setup Instructions

This tutorial provides comprehensive step-by-step directions for installing and operating the Face and Emotion Recognition Attendance System.

1. Requirements

Before installation, ensure your system meets the following prerequisites: Python version 3.8 or later must be installed along with pip, the Python package manager, and Ollama, which can be acquired and installed from https://ollama.com/.

2. Installation

Begin by cloning or downloading the project repository and unzipping all files into a designated folder on your computer. Next, open a command prompt or terminal and install the necessary models for Ollama by executing the commands ollama pull llama3.2 and ollama pull nomic-embed-text. Then, navigate to the project directory in your terminal and install all required Python packages using the command pip install streamlit opencv-python face-recognition dlib pandas numpy tensorflow Pillow langchain langchain-ollama langchain-chroma kagglehub, noting that the dlib installation can sometimes present difficulties and may require special operating system-specific instructions.

3. System Configuration

Execute the download_dataset.py script with the command python download_dataset.py to automatically download and install the FER emotion dataset. Then, run the train_emotion_model.py script using python train_emotion_model.py to build the essential CNN_Emotion_Model_7classes.h5 file, a process whose duration depends on your hardware capabilities. For student registration, create a folder named Photos in the main project directory and place clear, well-lit images of each student inside, ensuring each file is renamed to the student's name (e.g., Soma_Vashista.jpg). Additionally, create a CSV file named student_performance.csv in the main directory with the headers Name,Marks,Attendance,Recent_emotion and populate it with corresponding data for each

student, making certain the Name column entries exactly match their respective photo filenames without the extension.

4. Running the System

4.1. Main Application (Attendance and Emotion UI)

Open a terminal within the project directory and execute the command streamlit run app.py to launch the application, which will automatically load the login page in your web browser where you must enter the correct credentials (username: abhijeet). Utilize the sidebar to navigate system features including the "Auto-Timer Attendance" for automatic detection and marking, the "Add New Face" utility for student registration, the "View Records" section to display all attendance logs from the database, and the "Real-time Stream" for a live video feed showing face and emotion detection without attendance logging.

4.2. RAG Agent (Student Analysis)

Ensure Ollama is running in the background, then open a new terminal in the project directory and initiate the interactive chat session by entering the command python main.py. The script will first load the vector store from your student_performance.csv file, and once the "Enter your question" prompt appears, you can pose various queries regarding student well-being, such as inquiring about students exhibiting symptoms of sadness with poor attendance, requesting summaries of individual student performance, or identifying students with excellent grades but low attendance.

References

The development of this system draws upon foundational and contemporary research across several domains of computer science and machine learning. For face recognition, the work is built upon the Histograms of Oriented Gradients (HOG) feature descriptor by Dalal & Triggs (2005), implemented using the dlib-ml toolkit (King, 2009), and leverages the deep metric learning principles established by FaceNet (Schroff et al., 2015). The emotion detection module utilizes the FER-2013 dataset, the creation of which is detailed in the report by Goodfellow et al. (2013), and employs Convolutional Neural Networks, whose foundational architecture for image recognition was pioneered by Lecun et al. (1998). The analytical capability of the system is enhanced through Retrieval-Augmented Generation (RAG), a paradigm introduced by Lewis et al. (2020) for knowledge-intensive tasks. Furthermore, the implementation relies on critical software libraries including the OpenCV library for computer vision (Bradski, 2000), NumPy for array programming (Harris et al., 2020), TensorFlow as the machine learning framework (Abadi et al., 2016), and the LangChain (Chase, 2022) and Streamlit (Streamlit Inc., 2020) frameworks for building the RAG agent and web application, respectively.